

Alex Zeng

Mechanical Engineer · Boston, MA

☎ (978) 482-6082 | ✉ zeng.al@northeastern.edu | 🏠 a-zeng.github.io | 🌐 a-zeng

WORK EXPERIENCE

Mechanical Engineer II | Vicarious Surgical, Inc. - Waltham, MA

January 2022 - Present

- Owned a majority of the design of the distal portion of a novel, 9 DoF, tendon-drive robotic instrument, guiding its transition from single prototypes to dozens of pre-production units following processes.
- Developed and verified the rotary joint sensing design on the instrument by developing multilayer printed circuit stackups using photochemical etching, increasing joint sensing resolution and linearity.
- Led two design iterations of rotary joint assemblies, performing tolerance analyses, DFM optimization with vendors, and FEA simulations, and verifying performance through testing to design specifications.
- Redesigned instrument components for plastic or metal injection molding, significantly reducing BoM cost and improving manufacturability while maintaining performance.
- Designed and executed an automated test fixture to validate instrument strength and reliability, logging 300+ test runs to quantify mean time to failure (MTTF) and common failure modes.
- Diagnosed and resolved performance issues in pre-production instruments by conducting root cause analyses, supporting designs of experiments to test modifications, and implementing verified design improvements.
- Managed the releases of the 200+ part top-level assembly and BoM of the instrument in Arena QMS, completing the documentation of all parts and engineering drawings ahead of schedule.
- Collaborated with controls and embedded teams to develop an instrument tester that executes input trajectories on a loaded instrument while collecting position data for analysis.
- Automated reliability and calibration tests using Python scripts on Ubuntu CLI to control motorized force testers.
- Prototyped and calibrated a flexure-based force sensor to achieve more representative grasping force measurements than conventional load cells.
- Designed and programmed instrument subassembly test fixtures to isolate high-risk failure modes, enabling faster iteration using fewer parts.

Junior Machinist | Northeastern University Machine Shop - Boston, MA

September 2021 - May 2023

- Fabricated parts using CNC (HSMWorks) and manual machines to support over 100+ parts requests per semester.
- Designed and manufactured custom tooling holders to reduce setup times.
- Utilized multiple work coordinate systems (WCS) to reduce machining cycle time.

PROJECTS

Integration Lead | Northeastern University Mars Rover Team - Boston, MA

September 2019 - Present

- Collaborated with Northeastern students to build a 50kg Mars rover to compete in the University Rover Challenge.
- Implemented an imitative haptic controller for the rover arm shaped as a scaled-down model of the arm to increase operator intuition of arm location, resulting in significantly more dextrous control of the arm.
- Designed a sheet metal rover frame that increased internal surface area by 34% and decreased weight by 17%.

EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Mechanical Engineering · GPA: 3.90

September 2019 - May 2023

- Honors and Awards: Graduated summa cum laude, Dean's Scholarship, Dean's List (Fall 2019 – Spring 2023)
- Relevant Courses: ME Capstone, CFD, Robot Mechanics and Controls, ME Design, Controls, Materials

SKILLS

Engineering SolidWorks (CSWA), GD&T, Tolerance Analysis, SolidWorks Simulation, MATLAB, ANSYS Fluent
Fabrication 3D Printing (FDM & SLA), CNC (Mill & Lathe), Manual (Mill & Lathe), Waterjetting, Laser Cutting
Languages Python, MATLAB, HTML / CSS, PHP, C